Evaluating Adaptive Region Acquisition for Spatial Transcriptomics Reconstruction

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Abstract

Region-based spatial transcriptomics profiles only part of a tissue section, so the choice of regions to profile is a budgeted decision. Recent methods make this choice adaptively, conditioning each next region on the expression already measured. We test whether adaptivity helps under two objectives: reconstructing expression in the unprofiled regions, and capturing regions of high molecular heterogeneity (ROI detection). The evaluation uses a simulator built from 22 human HEST-1k slides (7 Visium HD, 15 Visium) from 18 studies and four tissues, a frozen UNI2-h encoder, a graph-based imputer, and strong non-adaptive placement baselines, including mutual-information-greedy sensor placement. Analyses were specified before the run, cluster by study, and correct for 24 primary comparisons. For reconstruction, adaptive policies were worse than the best non-adaptive baseline at budgets of 0.5–2% of regions (mean differences -0.128 to -0.040 PCC over the top-50 genes; Holm-adjusted $p < 0.001$) and within ± 0.01 PCC of it at 10–20%. A policy that selects directly on measured expression values was worse at every budget (-0.136 to -0.165). For ROI detection, the same policy raised recall by $+0.001$ to $+0.042$ over the best coverage baseline. None of these gains remained significant after correction (adjusted $p \geq 0.36$). They were concentrated in the Visium HD slides and were of similar size to the gain of a non-adaptive diversity sampler. On synthetic data with strong ROI signal, the protocol detected an adaptivity advantage of $+0.10$ recall, so the null results are not a property of the simulator alone. In these experiments, the main lever for both objectives was the placement rule, not adaptivity.

1 Introduction

Whole-transcriptome spatial assays are costly. Region-of-interest platforms such as GeoMx DSP [9] profile a small, operator-chosen subset of a section, and regions that are not profiled are reconstructed when needed, usually by predicting expression from co-registered hematoxylin and eosin (H&E) images [5, 11]. Which regions are profiled therefore affects two different things: how much of the section's expression can be recovered, and whether the regions that matter biologically are measured at all.

Several recent methods treat region choice as a sequential decision. SOFisher [7] trains a reinforcement-learning policy that moves fields of view toward regions enriched for a target signal, using the expression measured so far. SCR^2-ST [15] uses reinforcement learning to choose sampling locations for expression prediction, with a reward dominated by coverage of a single-cell reference manifold. S2-omics [14] selects ROIs from histology alone, before any measurement, to capture molecular heterogeneity, and notes that sequential designs that wait for measured expression are hard to run in practice. The two objectives in these papers differ: SOFisher and S2-omics aim to capture particular regions, whereas reconstruction aims to predict expression everywhere. An improvement under one objective need not carry over to the other.

Preprint. Under review.

We ask a narrow question under matched conditions. With the image encoder and imputer fixed, and against strong non-adaptive placement rules, does conditioning region choice on measured expression improve (i) whole-section reconstruction or (ii) ROI detection, and does any improvement survive acquisition in batches or contiguous blocks, as real platforms require? For reconstruction, Gaussian-process sensor placement gives a reason to expect little benefit: with a fixed kernel the posterior variance does not depend on the observed values, so adaptive and non-adaptive placement share the same optimum [6]. For detection, there is no such argument, since the target set is defined by the values themselves.

Contributions. (1) A simulator protocol for region acquisition on 22 human HEST-1k slides from 18 studies, with analyses that resample whole studies and correct for multiple comparisons, specified and committed before the expanded run. (2) For reconstruction, adaptive policies were worse than the best non-adaptive placement at low budgets and equivalent to it within 0.01 PCC at 10–20%; a policy that selects on measured values was worse at every budget. (3) For ROI detection, value-conditioned selection gave small recall gains that did not survive correction, were concentrated in one platform, and were matched by a non-adaptive diversity sampler. Synthetic positive controls show that the protocol can register an adaptivity advantage when one exists. (4) Code, configuration, and scripts that regenerate every table and figure from the result files.

2 Related work

Predicting expression from histology. Methods for predicting spatial expression from H&E include contrastive retrieval [11], flow matching over spatial neighborhoods [4], diffusion models [16], and autoregressive models [10], benchmarked on HEST-1k [5]. We hold the predictor fixed and study which regions it is given.

Choosing what to measure. Gene-panel selectors choose a fixed set of genes [2, 8]. Region selectors include one-shot rankers that score regions from images [12], histology-based ROI selection for molecular heterogeneity [14], and sequential reinforcement-learning samplers [7, 15]. These papers compare against random, uniform, or manual selection; we add sensor-placement baselines and evaluate both objectives on the same slides.

Active acquisition and sensor placement. Active feature acquisition treats measurement as a sequential policy [13]. Variance- and mutual-information-greedy placement in Gaussian processes are submodular and near-optimal for reconstructing a spatial field [6]; we use them as non-adaptive baselines.

3 Experimental setup

Slides and acquisition unit. We use 22 human slides from HEST-1k [5] across prostate (6), kidney (4), lung (6), and colon (6); 7 are Visium HD and 15 are standard Visium (Appendix A). Slides were selected by rules fixed in advance: human, whole-transcriptome, at least 1,000 spots under tissue, source image at most 0.6 μm/px, one slide per specimen, and at most two slides per study. The 22 slides come from 18 HEST studies. The acquisition unit ("region") is one native spot. For Visium, spots are 55 μm in diameter on a 100 μm pitch; the Visium HD slides are distributed by HEST as aggregated pseudo-spots with a median spacing of 128 μm on our slides. Coordinates are converted to microns with each slide's estimated pixel size. Slides contain 1,766 to 9,080 regions. Each region's image input is the HEST 224 px patch at 0.5 μm/px centred on the spot, embedded with frozen UNI2-h [1]. Expression is log1p-transformed; genes detected in fewer than 10% of regions are removed.

Simulator and objectives. A policy reveals regions up to a budget given as a fraction of the slide's regions (0.5% to 20%). A revealed region exposes its measured expression. For *reconstruction*, a fixed imputer predicts the unprofiled regions from the profiled ones, and we score the per-gene Pearson correlation over unprofiled regions, averaged over the top-50 highly variable genes (PCC), following the HEST-Benchmark target [5]. For *ROI detection*, we define ROIs from ground truth as the 20% of regions with the highest local molecular heterogeneity (variance of top-50-gene expression

across a region's 8 nearest spatial neighbours, averaged over genes), in the spirit of S2-omics [14], and score recall: the fraction of ROI regions that were profiled. ROIs and gene sets are used only for evaluation.

Imputer. The primary imputer (graphflow) maps image principal components through 512 random Fourier features, fits a ridge regression to expression on the profiled regions, and then runs 20 steps of message passing on a fixed 8-nearest-neighbour graph in image-and-space features, holding profiled regions at their measured values. Its predictions depend on the measured values. Its uncertainty, one minus the graph influence reaching a region from profiled nodes, depends on which regions are profiled but not on their values. A k-nearest neighbour and a regression-kriging imputer were run on the earlier slide set only (Appendix D).

Policies. Non-adaptive: random (5 seeds), farthest-point in space (grid), k-means on image embeddings with medoids, a reimplementation of a one-shot image ranker [12], and variance-greedy and mutual-information (MI)-greedy placement with an RBF kernel on joint image-and-coordinate features [6]. Sequential: *error-reduction* selects the region whose 15 nearest unprofiled neighbours in the imputer's feature space have the largest summed imputer uncertainty; *maximum-uncertainty* selects the region of highest imputer uncertainty. Under graphflow both depend on the profiled set but not on the measured values. *Value-conditioned* selection scores each candidate by the standardized kernel-weighted local variance of the measured expression around it plus 0.5 times its standardized distance to the nearest profiled region, and starts from the region nearest the slide centroid; its choices depend on the measured values, so it is the direct test of value-adaptivity. Two further policies implement the stated objectives of prior samplers in our simulator: a *diversity* sampler (farthest-point in joint image and space, after SCR^2-ST [15], non-adaptive) and an *ROI-seeking* sampler (moves toward measured regions with large expression deviation, after SOFisher [7], adaptive). These are reimplementations of the objectives, not the authors' trained models.

Comparators and statistics. For reconstruction, the comparator on each slide is the best of the five non-adaptive policies other than random ("best non-adaptive"). For detection and the realism experiments, it is the best of k-means, grid, and MI-greedy ("best coverage"). Taking the per-slide maximum favours the baseline. For each margin (policy minus comparator, per slide) we report the mean, a 95% percentile bootstrap CI that resamples whole studies (5,000 resamples), and an exact two-sided sign-flip test that flips the signs of whole studies (all 2^{18} patterns). Slides from the same study are therefore not treated as independent. Before the run we fixed a primary family of 24 margins (reconstruction and detection at each budget, and the realism settings) and adjust its p-values with Holm's method [3] (p_H). The decision rules, also fixed in advance, were: an adaptivity advantage is claimed only for a positive margin with $p_H < 0.05$.

4 Results

4.1 Synthetic controls: can the protocol register an adaptivity advantage?

We first checked that the simulator and metrics can show an advantage when one exists, using synthetic 20×20 slides with image-unpredictable, spatially clustered structure (8 slides per setting, 10% budget, k-NN imputer; Figure 1). For reconstruction, we added a non-deployable *difficulty oracle* that orders regions by true local heterogeneity weighted by distance to regions already chosen. It did not exceed the best coverage baseline at any structure length: its margin was -0.105 and -0.066 PCC at the two shortest correlation lengths and between -0.013 and -0.020 at the three longer ones, with every CI below zero (Figure 1a). For detection, the value-conditioned policy's recall margin over the best coverage baseline rose with the ROI signal amplitude α, from -0.045 at $\alpha = 0$ to $+0.052$ (95% CI $+0.033$ to $+0.071$) at $\alpha = 4$ and $+0.101$ ($+0.065$ to $+0.135$) at $\alpha = 8$, positive on all 8 slides in both settings (Figure 1b). The error-reduction policy, which targets the imputer's uncertainty (here the variance of measured expression among neighbours) rather than local heterogeneity, stayed at or below zero. The protocol can therefore register a value-adaptivity advantage for detection when the ROI signal is strong, and in this setting even ground-truth-informed ordering did not help reconstruction.

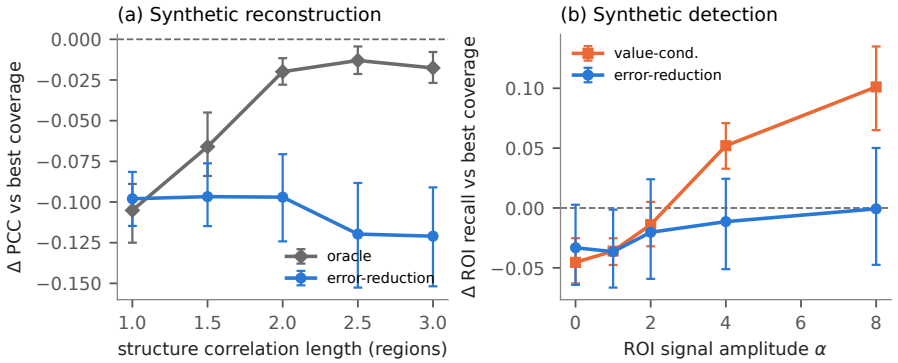


Figure 1: Synthetic controls (8 slides per point, 10% budget, mean and 95% bootstrap CI over slides). (a) Reconstruction margin over the best coverage baseline versus the correlation length of the hidden structure, for a ground-truth difficulty oracle and the error-reduction policy. (b) Detection-recall margin versus ROI signal amplitude. The dashed line marks zero.

Table 1: Absolute reconstruction accuracy (PCC over the top-50 genes on unprofiled regions), mean over 22 slides, graphflow imputer. "Best per slide" is the highest-scoring non-adaptive policy on each slide. Value-conditioned selection was run at 2–20% only.

	Non-adaptive			Adaptive		
Budget	best per slide	MI-greedy	k-means	error-red.	max-unc.	value-cond.
0.5%	0.470	0.412	0.449	0.343	0.348	–
1%	0.550	0.522	0.534	0.467	0.462	–
2%	0.607	0.599	0.598	0.567	0.562	0.471
5%	0.660	0.659	0.653	0.650	0.648	0.513
10%	0.687	0.685	0.679	0.684	0.685	0.535
20%	0.709	0.707	0.696	0.708	0.710	0.544

4.2 Reconstruction on 22 slides

Table 1 gives absolute PCC. The best non-adaptive baseline reached 0.470 at 0.5% budget and 0.709 at 20%; MI-greedy alone reached 0.412 and 0.707. Table 2 and Figure 2a give the paired margins. The error-reduction policy was below the best non-adaptive baseline by 0.128, 0.083, and 0.040 PCC at 0.5%, 1%, and 2% ($p_H < 0.001$; lower in all 18 studies at 0.5% and 1% and in 17 of 18 at 2%) and by 0.011 at 5% ($p_H = 0.057$). At 10% and 20% the margins were -0.002; their 90% study-cluster CIs, $[-0.006, +0.001]$ and $[-0.003, +0.001]$, lie within ± 0.01 PCC, so the two are equivalent at that tolerance (an equivalence margin we chose after seeing the data). The maximum-uncertainty policy behaved similarly; at 20% it was $+0.001$ above the best non-adaptive baseline (95% CI -0.001 to $+0.003$, unadjusted $p = 0.35$). The value-conditioned policy, the one that selects on measured values, was below the best coverage baseline at every budget, by 0.136 to 0.165 PCC ($p_H < 0.001$), reaching 0.544 at 20% against about 0.70 for the coverage policies.

4.3 ROI detection on 22 slides

Table 3 gives absolute recall. Coverage policies captured ROIs at about the rate expected from the budget (recall 0.200–0.232 at 20%), and the ground-truth ordering reached 0.999. The value-conditioned policy reached 0.278. Its margins over the best coverage baseline were $+0.001$, $+0.013$, $+0.021$, and $+0.042$ at 2%, 5%, 10%, and 20% (Table 2, Figure 2b). The unadjusted p-values at 5–20% were 0.040–0.044, and after Holm adjustment over the primary family all were 0.36. By the pre-specified rule, these margins do not establish a detection advantage.

Two further comparisons bear on how to read the small positive estimates. First, the non-adaptive diversity sampler improved recall over the best coverage baseline by a similar amount: $+0.020$ (95% CI $+0.009$ to $+0.031$) at 10% and $+0.029$ ($+0.018$ to $+0.042$) at 20%, with Holm-adjusted p of 0.015 and 0.002 within that experiment, where the value-conditioned margins had adjusted $p = 0.36$

Table 2: Primary margins over 22 slides (18 study clusters). Δ is the mean per-slide difference with a 95% bootstrap CI that resamples studies; p_H is the exact study-level sign-flip p-value, Holm-adjusted over the 24 pre-specified primary margins (this table and Table 4).

	Reconstruction (PCC, top-50 genes)				Detection (ROI recall)	
	error-reduction − best non-adapt.		value-cond. − best coverage		value-cond. − best coverage	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p_H
0.5%	$-0.128\ [-0.162, -0.096]$	<0.001	–	–	–	–
1%	$-0.083\ [-0.105, -0.064]$	<0.001	–	–	–	–
2%	$-0.040\ [-0.054, -0.027]$	<0.001	$-0.136\ [-0.171, -0.106]$	<0.001	$+0.001\ [-0.004, +0.006]$	0.723
5%	$-0.011\ [-0.018, -0.004]$	0.057	$-0.148\ [-0.180, -0.119]$	<0.001	$+0.013\ [+0.002, +0.024]$	0.362
10%	$-0.002\ [-0.007, +0.001]$	0.687	$-0.152\ [-0.186, -0.122]$	<0.001	$+0.021\ [+0.003, +0.037]$	0.362
20%	$-0.002\ [-0.004, +0.001]$	0.687	$-0.165\ [-0.217, -0.125]$	<0.001	$+0.042\ [+0.006, +0.077]$	0.362

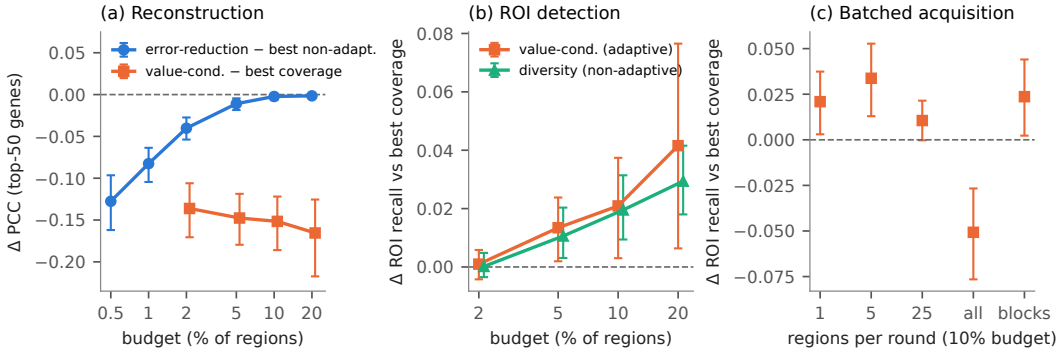


Figure 2: Margins over 22 slides, mean and 95% CI resampling studies. (a) Reconstruction. (b) ROI-detection recall for the value-conditioned policy and the non-adaptive diversity sampler. (c) Detection margin of the value-conditioned policy at 10% budget as acquisition is batched (regions per round; "all" is a single round) or made contiguous ("blocks": each round reveals the 7 regions nearest a chosen seed).

(Table 3, Figure 2b). Second, the detection margins were concentrated in the Visium HD slides: at 10% and 20% they were $+0.048$ and $+0.099$ on HD (7 slides, 5 studies, the smallest attainable p with 5 clusters is 0.0625) and $+0.008$ and $+0.015$ on Visium (15 slides, 13 studies; $p = 0.43$ and 0.47; Appendix B). The ROI-seeking sampler did not improve recall (-0.012 to -0.005, all CIs including zero) and reduced reconstruction PCC by 0.19 to 0.28.

4.4 Batched and contiguous acquisition

Real platforms acquire regions in rounds, and ROIs are contiguous. At 10% budget we revealed regions in rounds of 1, 5, or 25, or in a single round, with the policy using only measurements available at the start of each round, and separately in contiguous blocks of 7 regions around a policy-chosen seed (Table 4, Figure 2c). Reconstruction margins were negative in every setting ($p_H < 0.001$). Detection margins were $+0.021$, $+0.034$, and $+0.011$ for rounds of 1, 5, and 25, and $+0.024$ for contiguous blocks, none with $p_H < 0.05$ (rounds of 5: $p_H = 0.073$). With a single round the policy has no measurements and falls back to its starting rule, which takes the regions nearest the slide centroid; this compact block captured fewer ROIs than the coverage policies (margin -0.051, $p_H = 0.008$).

4.5 Robustness

Gene set. Selecting the top-50 genes from the full slide uses information that a prospective experiment would not have. Selecting them instead from a random 15% reference subset gave reconstruction margins within 0.003 of the full-slide choice, and scoring all filtered genes gave smaller but still negative margins (-0.033 to -0.079; all CIs below zero; Appendix C). *Tissue.* The value-conditioned reconstruction margins were negative in all four tissues at every budget, as were the error-reduction margins at 0.5–5%; at 10–20% the error-reduction margin in kidney was $+0.001$

Table 3: Absolute ROI-detection recall (fraction of the top-20% most heterogeneous regions profiled), mean over 22 slides. †Reimplementations of the stated objectives of SCR^2-ST (diversity, non-adaptive) and SOFisher (ROI-seeking, adaptive) in our simulator. *Profiles regions in order of true heterogeneity; it uses ground truth and bounds recall from above.

	Non-adaptive coverage			Competitor objectives			
Budget	k-means	grid	MI-greedy	diversity†	ROI-seeking†	value-cond.	oracle*
2%	0.018	0.022	0.018	0.026	0.021	0.027	0.100
5%	0.050	0.051	0.046	0.067	0.050	0.070	0.250
10%	0.110	0.103	0.094	0.136	0.104	0.137	0.500
20%	0.232	0.204	0.200	0.266	0.232	0.278	0.999

Table 4: Value-conditioned policy minus best coverage at 10% budget under batched and contiguous acquisition (22 slides; Δ with 95% CI resampling studies; p_H Holm-adjusted over the primary family).

	Reconstruction (PCC)		Detection (ROI recall)	
Acquisition setting	Δ [95% CI]	p_H	Δ [95% CI]	p_H
Sequential (1 region per round)	-0.152 $[-0.186, -0.122]$	<0.001	$+0.021$ $[+0.003, +0.037]$	0.362
Rounds of 5 regions	-0.129 $[-0.175, -0.094]$	<0.001	$+0.034$ $[+0.013, +0.053]$	0.073
Rounds of 25 regions	-0.130 $[-0.176, -0.095]$	<0.001	$+0.011$ $[-0.000, +0.021]$	0.362
Single round (whole budget)	-0.236 $[-0.288, -0.188]$	<0.001	-0.051 $[-0.077, -0.027]$	0.008
Contiguous blocks of 7 regions	-0.087 $[-0.116, -0.063]$	<0.001	$+0.024$ $[+0.002, +0.044]$	0.362

and $+0.000$. The detection margins were positive in kidney, lung, and colon and between -0.006 and -0.001 in prostate at 2–10% (Appendix B). *Native SOFisher.* We ran the authors' SOFisher code on its bundled mouse cortex data for 5 million steps; the evaluation reward for captured target regions rose from a median of -238 early in training to $+147$ late (Appendix E), confirming that the method optimizes target-region capture. We could not run SCR^2-ST natively because its processed training data are not publicly available.

5 Discussion

Reconstruction. The reconstruction results agree with the sensor-placement argument. With the imputer's uncertainty independent of measured values, the sequential policies are in effect alternative fixed designs, and they matched the best fixed design at moderate budgets and fell behind it at low budgets, where coverage matters most. Selecting on measured values moved measurements toward locally variable areas and away from coverage, which lowered PCC by 0.14 to 0.17. On synthetic data, ordering by true local difficulty did not help either. Whole-section reconstruction rewards spreading measurements across the section, and adaptivity in these experiments only redistributed measurements away from that.

Detection. Detection is the objective under which adaptivity could help, and the synthetic control shows that it does when the ROI signal in measured expression is strong. On real slides the estimated gain was at most 0.04 recall at 20% budget, against a gap of about 0.72 to the ground-truth ordering, and it did not survive correction. A non-adaptive diversity design gave a similar gain, so the part of the gain we can resolve is not specific to adaptivity. The concentration in Visium HD slides may reflect the smoother pseudo-spot signal or the particular studies; with 5 HD studies we cannot separate these.

Practical implication. For budgets and slides like these, a placement rule computed in advance (MI-greedy or k-means) was as accurate for reconstruction as any adaptive policy we tested. For detection, a non-adaptive diversity design reached recall within 0.012 of the value-conditioned policy (0.266 against 0.278 at 20% budget) without waiting for measurements between rounds.

Limitations. The simulator is retrospective: a revealed region returns its measured expression without additional noise, batch effects, or turnaround time, which favours adaptive policies. The

acquisition unit is a spot or HD pseudo-spot, smaller than typical GeoMx ROIs, and the HD and Visium units differ in spacing. The 22-slide results use one imputer; the kNN and kriging comparisons rest on 5 slides. The ROI definition is one choice among several, and ROIs from expert annotation could behave differently. The prior samplers were evaluated as reimplementations of their objectives in our simulator, not as the authors' trained systems; SOFisher was run natively only on its own data, and SCR^2-ST not at all. The equivalence tolerance of 0.01 PCC was chosen after the data were seen. Budgets above 20% were not tested.

6 Conclusion

Across 22 human slides from 18 studies, adaptive region acquisition did not improve whole-section reconstruction over strong non-adaptive placement: it was worse at low budgets and equivalent within 0.01 PCC at 10–20%, and selecting on measured values was worse at every budget. For ROI detection, value-conditioned selection produced small recall gains that did not survive correction for multiple comparisons and were matched by a non-adaptive diversity design. The same protocol registered a clear detection advantage on synthetic data with strong ROI signal, which marks the setting in which adaptivity is worth testing again: targets that are invisible in histology but strongly expressed once measured.

Reproducibility. Code, configuration, result files, and the scripts that regenerate every table and figure (`scripts/14_power_summary.py`, `scripts/15_paper_assets.py`) are available at https://github.com/ayushdeo/STARE.

A Slides

Table 5 lists the slides. "Cluster" indexes the HEST study used as the resampling unit; "Pitch" is the median nearest-neighbour spacing of regions in microns.

Table 5: The 22 slides (HEST-1k v1.3 metadata).

Tissue	HEST ID	Platform	Spots	μm/px	Pitch (μm)	Cluster	Study (HEST dataset title)
prostate	TENX174	Visium HD	2385	0.274	128	1	Visium HD Spatial Gene Expression Library, Human Prostate ...
prostate	TENX40	Visium	4371	0.345	100	2	Human Prostate Cancer, Adenocarcinoma with Invasive Carcin...
prostate	MEND160	Visium	4079	0.573	100	3	Spatially resolved clonal copy number alterations in benig...
prostate	TENX46	Visium	3043	0.344	100	4	Human Prostate Cancer, Acinar Cell Carcinoma (FFPE)
prostate	TENX41	Visium	2543	0.345	100	5	Normal Human Prostate (FFPE)
prostate	MEND144	Visium	4074	0.341	100	3	Spatially resolved clonal copy number alterations in benig...
kidney	TENX173	Visium HD	2684	0.274	128	6	Visium HD Spatial Gene Expression Library, Human Kidney (F...
kidney	TENX71	Visium	5936	0.274	100	7	Human Kidney, 11 mm Capture Area (FFPE)
kidney	INT21	Visium	4975	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
kidney	INT19	Visium	4948	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
lung	TENX171	Visium HD	1766	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX170	Visium HD	1877	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX163	Visium HD	2487	0.274	128	10	Visium HD Spatial Gene Expression Library, Human Lung Canc...
lung	TENX72	Visium	6195	0.274	100	11	Human Lung Cancer, 11 mm Capture Area (FFPE)
lung	TENX62	Visium	3858	0.273	100	12	Human Lung Cancer (FFPE)
lung	MISC20	Visium	3316	0.438	100	13	A spatially resolved atlas of the human lung characterizes...
colon	TENX156	Visium HD	2179	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX155	Visium HD	2281	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX70	Visium	9080	0.274	100	15	Human Colorectal Cancer, 11 mm Capture Area (FFPE)
colon	TENX92	Visium	6518	0.274	100	16	Visium CytAssist Gene Expression Libraries of Post-Xenium ...
colon	MISC47	Visium	4818	0.276	100	17	COLON MAP: Colon Molecular Atlas Project
colon	ZEN48	Visium	2385	0.459	100	18	Charting the Heterogeneity of Colorectal Cancer Consensus ...

Changes from the earlier seven-slide analysis. An earlier version of this analysis used seven Visium HD slides and described regions as 112 μm squares. Two corrections apply. First, the current HEST metadata lists two of those slides as non-human (TENX186, *Mus musculus*; TENX175, mixed human and mouse); both are excluded here. Second, the loader had assumed 0.5 μm/px, whereas the HD images are about 0.274 μm/px, so the regions were about 61 μm squares containing exactly one pseudo-spot each. The present analysis defines the unit as the spot directly. On the five retained slides, the per-slide value-conditioned margins (reconstruction and detection, four budgets) differ between the two runs by a median of 0.015 (maximum 0.096), from the use of true spot positions instead of grid centres; 3 of 40 change sign, two of them smaller than 0.005 in magnitude and one detection margin moving from -0.050 to $+0.015$. With seven slides the smallest attainable exact p-value is 0.016, so none of the earlier margins could have survived a correction for 24 comparisons.

B Platform and tissue splits

Table 6: Selected margins by platform (with unadjusted study-level sign-flip p) and by tissue (mean). With 5 HD studies the smallest attainable p is 0.0625.

	HD (n=7, 5 cl.)	Visium (n=15, 13 cl.)	prostate (n=6)	kidney (n=4)	lung (n=6)	colon (n=6)
Recon., error-red. (1%)	-0.113 (0.062)	-0.068 ($<$0.001)	-0.078	-0.085	-0.081	-0.088
Recon., value-cond. (10%)	-0.103 (0.062)	-0.174 ($<$0.001)	-0.115	-0.171	-0.141	-0.185
Detect., value-cond. (10%)	$+0.048$ (0.062)	$+0.008$ (0.426)	-0.001	$+0.022$	$+0.036$	$+0.027$
Detect., value-cond. (20%)	$+0.099$ (0.062)	$+0.015$ (0.469)	$+0.015$	$+0.024$	$+0.052$	$+0.070$
Detect., rounds of 5 (10%)	$+0.049$ (0.250)	$+0.026$ (0.044)	$+0.016$	$+0.012$	$+0.046$	$+0.053$

C Gene-set sensitivity

Table 7: Value-conditioned minus best coverage reconstruction margin (PCC) under three gene sets, 22 slides, 95% CI resampling studies.

Budget	per-slide top-50 HVG	reference-subset top-50 HVG	all filtered genes
2%	-0.136 $[-0.171, -0.106]$	-0.134 $[-0.168, -0.103]$	-0.032 $[-0.047, -0.021]$
5%	-0.148 $[-0.180, -0.119]$	-0.148 $[-0.182, -0.118]$	-0.052 $[-0.068, -0.040]$
10%	-0.152 $[-0.186, -0.122]$	-0.153 $[-0.189, -0.122]$	-0.068 $[-0.087, -0.052]$
20%	-0.165 $[-0.217, -0.125]$	-0.166 $[-0.217, -0.126]$	-0.079 $[-0.101, -0.061]$

D Imputer comparison on the earlier slide set

The kNN and kriging imputers were run only on the earlier slide set. Table 8 reports the error-reduction policy minus the best non-adaptive baseline on the five human slides from that run (legacy unit, which equals one pseudo-spot per region on these slides), with the number of slides on which the margin was positive. With five slides these values are descriptive.

Table 8: Error-reduction minus best non-adaptive (PCC), mean over 5 human Visium HD slides (positive/total).

Budget	kNN	kriging	graphflow
0.5%	-0.088 (0/5)	-0.088 (2/5)	-0.082 (0/5)
1%	-0.114 (0/5)	-0.138 (1/5)	-0.083 (0/5)
2%	-0.091 (0/5)	-0.150 (1/5)	-0.040 (0/5)
5%	-0.121 (0/5)	-0.141 (1/5)	-0.011 (1/5)
10%	-0.134 (0/5)	-0.133 (0/5)	-0.000 (2/5)
20%	-0.145 (0/5)	-0.126 (0/5)	-0.002 (2/5)

E Native SOFisher run

We ran SOFisher from the authors' repository on its bundled mouse cortex data with default settings for 5 million steps (seed 222). The only change was the addition of checkpoint and resume support, so that the run could be continued after interruptions. Figure 3 shows the evaluation reward, which counts captured target regions. Its median over the first 5% of training was -238 and over the last 10% was $+147$ (best 514). Rare evaluation episodes have rewards near -1.7×10^5, so we plot rolling medians.

F Configuration

All values are fixed in the released `config.py`. Seeds: global 2021; random policy $\{0, \ldots, 4\}$. Imputer (graphflow): image PCA 64, random Fourier features 512 (median-heuristic bandwidth), ridge 10, graph $k = 8$, 20 steps, step size 0.5. kNN imputer: $k = 10$, joint image and space

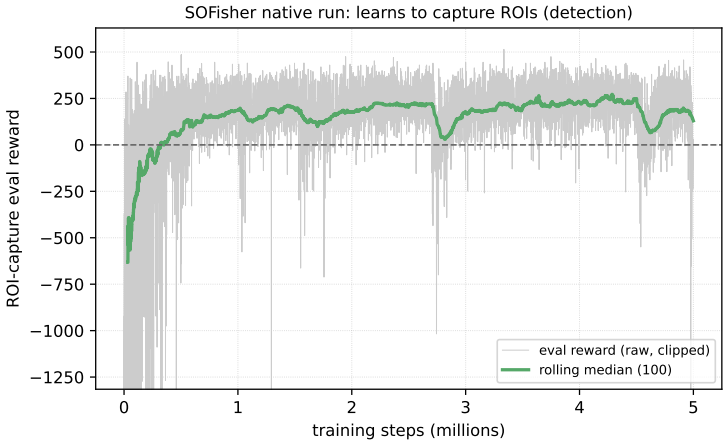


Figure 3: SOFisher evaluation reward during training (rolling median), native run.

features. Kriging: image PCA 64, ridge 10, residual $k = 10$. Placement: feature PCA 64, median-heuristic lengthscale, noise 10^{-2}. Error-reduction neighbourhood 15. Value-conditioned: spatial kernel width 0.1 of the slide diagonal, exploration weight 0.5; the ROI-seeking sampler uses the same values. Detection ROIs: top 20% by heterogeneity over 8 spatial neighbours. Statistics: 5,000 bootstrap resamples, exact sign-flip enumeration over study clusters, Holm over 24 primary margins. The code used for the larger slides replaces dense distance computations with sparse and incremental ones; on a 132-run check (11 policies, 3 imputers, 2 slide sizes, 2 budgets) it reproduced the original code's selections and scores exactly.

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